

B-EPD .BE
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Aliplast NV
STAR 75 double casement
& double casement turn and
titled
window system

aliplast[®]
ALUMINIUM SYSTEMS

ISSUED 04.07.2025
VALID UNTIL 04.07.2030

THIRD PARTY VERIFIED
in accordance with EN 15804+A2
and B-EPD PCR

FUNCTIONAL UNIT AND MODULES DECLARED

Coverage of 1 m² of external wall with a window (with a surface of < 2.3 m²) providing thermal and acoustic insulation, natural light transmission, water and air tightness and natural ventilation over a lifespan of 60 years.
Cradle-to-gate with options + C1-C4 and D

A123	A4	A5	B	C	D
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The intended use of this EPD is to communicate scientifically based environmental information for construction products, for the purpose of assessing the environmental performance of buildings. This EPD is only valid when registered on www.b-epd.be. The FPS Public Health cannot be held responsible for the information provided by the owner of the EPD.

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1 PRODUCT DESCRIPTION

1.1 Product name

Star 75 double casement & double casement turn-and-tilled aluminium profile window systems with a surface area of less than 2.3 m².

1.2 Product description and intended use

Star 75 double casement & Star 75 double casement turn-and-tilled are aluminium window systems. The frames are made of aluminium profiles with thermal breaks. Elastomer seals are used to ensure watertightness.

This is a specific EPD for a specific product produced by Aliplast NV.

Intended use:

- Window system is used in a window in all types of residential and non-residential buildings, both for new-build and renovations.

1.3 Reference flow / declared unit

COVERAGE OF 1 M² OF EXTERNAL WALL WITH A WINDOW (WITH A SURFACE OF < 2.3 M²) PROVIDING THERMAL AND ACOUSTIC INSULATION, NATURAL LIGHT TRANSMISSION, WATER AND AIR TIGHTNESS AND NATURAL VENTILATION OVER A LIFESPAN OF 60-YEARS.

Packaging is included.

The weight per reference flow is 18.64 kg.

The reference flow is based on the representative product – Double casement turn-and-tilled window system.



1.4 Installation

The scope of this product is 'AS PRODUCED'.

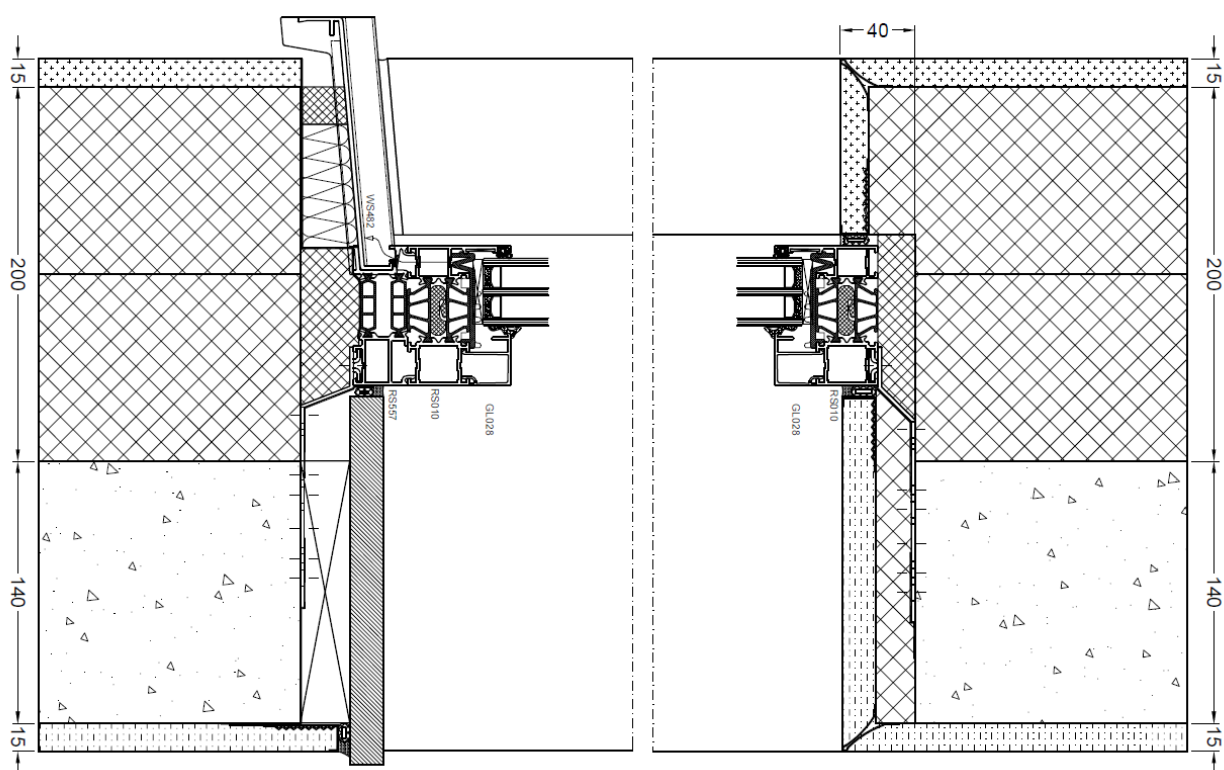
Materials for fixation and installation (glass) are not included.

This EPD only includes the environmental impact related to the product itself and end-of-life treatment of packaging.

For installing the product, one should follow the instructions provided by Aliplast N.V. The following scenarios are possible:

- Mounting brackets
- - Fixation screws.

This may lead to the need of additional products and materials for which the impact is not included in this EPD and which shall be taken into account at building level. More detailed information on these scenarios can be found in the chapter “Additional technical information for scenario development at building”.



1.5 Composition and content

Components	Composition	Quantity, kg
Product	– Aluminium	– 13.1
	– Metal parts	– 1.32
	– Rubber	– 1.44
	– Plastic Parts	– 2.11
	– Sealants, glues, coatings, etc	– 0.68
Packaging	– Cardboard	– 2.37E-01
	– Wood	– 2.68E-01
	– LDPE film	– 1.01E+00
	– PE straps	– 1.62E-02

The product does not contain materials listed in the “Candidate list of Substances of Very High Concern for authorization”.

1.6 Reference service life

The reference service life is estimated at 60 years.

Aliplast provides the necessary documentation on the expected performance, installation and maintenance instructions, as well as a list of spare parts and accessories in the product catalog and according to ATG 3008.

The conditions under which this RSL is valid are defined in the [STAR 75 catalogues](#) and ATG 3008.

In normal use, aluminium building products are not altered or corroded over time. A regular cleaning (e.g. once a year) of the product suffices to secure a long service life. However, the use of highly alkaline (pH >10) or highly acidic (pH < 4) cleaning solutions should be avoided. In practice, a service life of 60 years can be assumed in normal use for such application, except for the IGU (Insulated Glass Unit) which needs to be replaced usually after 30 years due to slow degradation of its performance.

1.7 Description of geographical representativity

The EPD is representative of a product produced, installed and discarded in Belgium.

The EPD is representative for the Belgian market.



1.8 Description of the production process and technology

The assembly process is generally carried out on site.

The following operations are carried out to produce the main parts:

1. Preparation of the aluminium profile (powder coating), mainly by sawing and milling.
2. Production of the frame by assembling the various profiles using connectors and fixing with bolts or glue. The connectors used are mainly made of aluminium.
3. Positioning and fixing the various joints.
4. Application of the infill (e.g. glazing, opaque panels).
5. Integration of equipment (where applicable).

2 TECHNICAL DATA / PHYSICAL CHARACTERISTICS

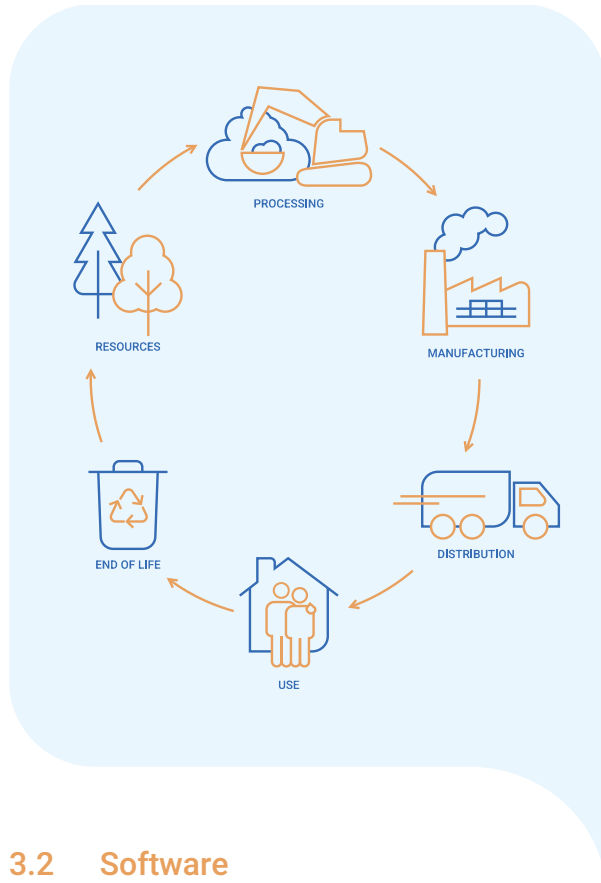
Technical property	Standard	Value	Unit	Comment
Dimensions	metric	1.23 x 1.48	m	
Thermal insulation (mandatory)	EN ISO 10077-2	0.94	W/m ² K	
Acoustic insulation	EN ISO 140-3 EN ISO 717-1	till 47.1	dB (-1,-4)	
Air tightness	EN 1026 EN12207	C5		
Fire resistance	EN 13501-2	EW30-EI30		
Water tightness	EN 1027 EN 12208	E1200A		



3 LCA-STUDY

3.1 Date of LCA-study

The LCA study has been performed in October 2024.



3.2 Software

For the calculation of the LCA results, the software program SimaPro 9.5.1 has been used.

3.3 Information on allocation

Allocation of inputs and outputs has been necessary in the manufacturing stage A3. Allocation has been performed using physical relationship – annual usage vs. annual volume of the specific window system produced. The allocation procedure described by the European standard EN 15804+A2 has been followed.

3.4 Information on cut-off

The following processes are considered below cut-off:

- Recycling of raw materials packaging.
- All activities and energy use related to the offices and personnel.
- Steel cassettes as packaging of one of the raw materials, as these are re-used countless times.
- Infrastructure & impacts related to the use of the machines at Aliplast's site.

Cut-off criteria described in EN15804+A2 has been followed: 5% of energy and materials by module and 1% of total energy and raw materials.

3.5 Information on excluded processes

The following processes were excluded from the inventory:

- Flows related to human activities such as employee transport and administration activity.



3.6 Information on biogenic carbon modelling

Only product packaging contains biogenic carbon. Following EN15804+A2, flows of biogenic carbon have been allocated according to physical properties, because the net sum of biogenic CO₂ uptake and release over modules A1-C4 should be zero. Relevant data records have been adjusted accordingly.

Biogenic carbon content	(kg C / FU)
Biogenic carbon content in product (at the gate)	0
Biogenic carbon content in accompanying packaging (at the gate)	2.39E-01

3.7 Information on carbon off-setting

Carbon offsetting is not allowed in EN 15804+A2 and hence not taken into account in the calculations.

3.8 Additional or deviating characterization factors

The characterization factors from EC-JRC were applied: EF 3.1. No additional or deviating characterization factors were used.

3.9 Description of the variability

This EPD covers 2 types of window systems. The differences between these systems are minor - quantities of only some components vary (handles, fittings, one plastic part). Consequently, this EPD is representative of the worst-case scenario.



3.10 Specificity

The data used for the LCA is specific for this product which is manufactured by a single manufacturer in a single production site.

3.11 Period of data collection

Manufacturer specific data have been collected for the year 2022.

3.12 Information on data collection

Foreground processes include:

- Raw Materials used, their packaging, transportation mode and related quantities,
- Energy mix and amount used for production,
- Types of product packaging and related quantities,
- Environmental impacts of raw materials for which EPDs were available,
- All other quantitative information.

Generic datasets were used for:

- Extraction and manufacturing of raw materials, packaging,
 - Transportation and treatment of waste from raw material packaging and production,
 - Waste treatment processes,
 - Transport to end-of-life treatment,
 - End-of-life treatment processes and scenarios.
- Declared value is representative of the representative product – Double casement turn-and-tilled Window System, which is also a worst case (highest impacts) of the product group covered by this EPD.

3.13 Database used for background data

Databases used for modelling include:

- Ecoinvent 3.9
- Industry 2.0

3.14 Energy mix

Aliplast purchases renewable energy mix from their supplier. Renewable energy mix was modelled based on the statistics for Belgium.

GWP of this mix is 0.049 kg CO₂ eq. / kWh.



4 PRODUCTION SITES

Alplast production site is located at Waaslandlaan 15, 9160 Lokeren, Belgium.

5 SYSTEM BOUNDARIES

Product stage			Construction installation stage		Use stage							End of life stage				Beyond the system boundaries
Raw materials	Transport	Manufacturing	Transport	Construction installation stage	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	De-construction demolition	Transport	Waste processing	Disposal	Reuse-Recovery-Recycling-potential
A1	A2	A3	A4	A5	B1	B2	B3	B4	B5	B6	B7	C1	C2	C3	C4	D
☒	☒	☒	☒	☒	☐	☐	☐	☐	☐	☐	☐	☒	☒	☒	☒	☒

X = included in the EPD

☐ = module not declared



6 POTENTIAL ENVIRONMENTAL IMPACTS PER REFERENCE FLOW

	Production			Construction process stage		Use stage							End-of-life stage				D Reuse, recovery, recycling
	A1 Raw material	A2 Transport	A3 manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
 GWP total (kg CO2 equiv/FU)	1,557E+02	7,285E-01	5,906E+00	3,730E-01	2,856E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	5,79E-01	3,23E+00	1,07E+01	-8,31E+01
 GWP fossil (kg CO2 equiv/FU)	1,554E+02	7,279E-01	5,579E+00	3,727E-01	2,165E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	5,79E-01	3,21E+00	1,07E+01	-8,13E+01
 GWP biogenic (kg CO2 equiv/FU)	-8,413E-01	2,360E-04	3,189E-01	1,201E-04	6,903E-01	ND	ND	ND	ND	ND	ND	ND	0,00E+00	1,84E-04	1,40E-02	8,25E-04	-7,26E-01
 GWP luluc (kg CO2 equiv/FU)	1,855E+00	3,685E-04	8,110E-03	1,840E-04	4,027E-04	ND	ND	ND	ND	ND	ND	ND	0,00E+00	2,81E-04	1,67E-03	2,26E-04	-1,05E+00
 ODP (kg CFC 11 equiv/FU)	9,848E-06	1,594E-08	1,712E-07	8,114E-09	7,090E-09	ND	ND	ND	ND	ND	ND	ND	0,00E+00	1,26E-08	3,54E-08	4,21E-08	-4,02E-06
 AP (mol H+ eq/FU)	1,262E+00	3,412E-03	1,766E-02	8,148E-04	1,901E-03	ND	ND	ND	ND	ND	ND	ND	0,00E+00	1,89E-03	1,15E-02	2,67E-03	-4,55E-01
 EP - freshwater (kg P equiv/FU)	5,583E-03	2,033E-01	2,202E+00	7,747E-01	6,496E-03	ND	ND	ND	ND	ND	ND	ND	3,92E+02	4,63E-06	8,13E-05	8,34E-06	-2,46E-03
 EP - marine (kg N equiv/FU)	2,033E-01	8,719E-04	3,926E-03	2,006E-04	6,084E-04	ND	ND	ND	ND	ND	ND	ND	0,00E+00	6,41E-04	1,32E-03	9,60E-04	-5,72E-02
 EP - terrestrial (mol N equiv/FU)	2,202E+00	9,400E-03	4,047E-02	2,089E-03	6,593E-03	ND	ND	ND	ND	ND	ND	ND	0,00E+00	6,86E-03	1,60E-02	1,05E-02	-6,30E-01
 POCP (kg Ethene equiv/FU)	7,747E-01	3,939E-03	1,881E-02	1,265E-03	2,372E-03	ND	ND	ND	ND	ND	ND	ND	0,00E+00	2,82E-03	5,44E-03	2,80E-03	-2,85E-01
 ADP Elements (kg Sb equiv/FU)	6,496E-03	1,981E-06	1,925E-05	1,218E-06	2,220E-06	ND	ND	ND	ND	ND	ND	ND	0,00E+00	1,86E-06	5,77E-05	3,28E-06	0,00E+00
 ADP fossil fuels (MJ/FU)	2,202E+03	1,072E+01	1,229E+02	5,294E+00	6,344E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	8,21E+00	1,98E+01	2,76E+00	-9,21E+02
 WDP (m³ water eq deprived /FU)	3,981E+01	4,738E-02	1,574E+00	2,183E-02	7,672E-02	ND	ND	ND	ND	ND	ND	ND	0,00E+00	3,34E-02	2,44E-01	1,21E-01	-1,46E+00

GWP TOTAL = TOTAL GLOBAL WARMING POTENTIAL (CLIMATE CHANGE); GWP-LULUC = GLOBAL WARMING POTENTIAL (CLIMATE CHANGE) LAND USE AND LAND USE CHANGE; ODP = OZONE DEPLETION POTENTIAL; AP = ACIDIFICATION POTENTIAL FOR SOIL AND WATER; EP = EUTROPHICATION POTENTIAL; POCP = PHOTOCHEMICAL OZONE CREATION; ADPE = ABIOTIC DEPLETION POTENTIAL – ELEMENTS; ADPF = ABIOTIC DEPLETION POTENTIAL – FOSSIL FUELS; WDP = WATER USE (WATER (USER) DEPRIVATION POTENTIAL, DEPRIVATION-WEIGHTED WATER CONSUMPTION)

7 RESOURCE USE

	Production			Construction process stage		Use stage							End-of-life stage				
	A1 Raw material	A2 Transport	A3 manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	D Reuse, recovery, recycling
PERE (MJ/FU, net calorific value)	2,21E+02	1,54E-01	1,60E+01	0,00E+00	7,66E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	2,10E-01	2,99E+00	2,84E-01	0,00E+00
PERM (MJ/FU, net calorific value)	1,92E+01	0,00E+00	-1,17E+01	0,00E+00	-7,53E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
PERT (MJ/FU, net calorific value)	2,32E+02	1,54E-01	1,27E+01	0,00E+00	1,25E-01	ND	ND	ND	ND	ND	ND	ND	0,00E+00	2,10E-01	2,99E+00	2,84E-01	0,00E+00
PENRE (MJ/FU, net calorific value)	1,03E+03	1,07E+01	8,03E+01	0,00E+00	3,24E+01	ND	ND	ND	ND	ND	ND	ND	0,00E+00	1,38E+01	2,00E+01	9,81E+01	0,00E+00
PENRM (MJ/FU, net calorific value)	1,06E+02	0,00E+00	3,90E+01	0,00E+00	-4,13E+01	ND	ND	ND	ND	ND	ND	ND	0,00E+00	0,00E+00	0,00E+00	-9,53E+01	0,00E+00
PENRT (MJ/FU, net calorific value)	1,13E+03	1,07E+01	1,22E+02	0,00E+00	6,34E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	1,38E+01	2,00E+01	2,76E+00	0,00E+00
SM (kg/FU)	1,70E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
RSF (MJ/FU, net calorific value)	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
NRSF (MJ/FU, net calorific value)	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
FW (m³ water eq/FU)	2,14E+00	1,50E-03	3,75E-02	0,00E+00	2,89E-01	ND	ND	ND	ND	ND	ND	ND	0,00E+00	1,84E-03	1,33E-02	8,81E-03	0,00E+00

PERE = USE OF RENEWABLE PRIMARY ENERGY EXCLUDING RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PERM = USE OF RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PERT = TOTAL USE OF RENEWABLE PRIMARY ENERGY RESOURCES; PENRE = USE OF NON-RENEWABLE PRIMARY ENERGY EXCLUDING NON-RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PENRM = USE OF NON-RENEWABLE PRIMARY ENERGY RESOURCES USED AS RAW MATERIALS; PENRT = TOTAL USE OF NON-RENEWABLE PRIMARY ENERGY RESOURCES; SM = USE OF SECONDARY MATERIAL; RSF = USE OF RENEWABLE SECONDARY FUELS; NRSF = USE OF NON-RENEWABLE SECONDARY FUELS; FW = NET USE OF FRESH WATER

8 WASTE CATEGORIES & OUTPUT FLOWS

	Production			Construction process stage		Use stage							End-of-life stage				D Reuse, recovery, recycling
	A1 Raw material	A2 Transport	A3 manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	
Hazardous waste disposed (kg/FU)	1,10E-01	6,60E-05	2,43E-04	0,00E+00	3,24E-03	ND	ND	ND	ND	ND	ND	ND	0,00E+00	8,76E-05	8,31E-02	3,47E-03	0,00E+00
Non-hazardous waste disposed (kg/FU)	8,38E+00	7,85E-01	4,46E-01	0,00E+00	7,26E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	6,72E-01	1,84E+00	3,14E-01	0,00E+00
Radioactive waste disposed (kg/FU)	1,85E-03	3,30E-06	8,98E-05	0,00E+00	5,80E-04	ND	ND	ND	ND	ND	ND	ND	0,00E+00	4,47E-06	5,54E-05	6,37E-06	0,00E+00
Components for re-use (kg/FU)	0,00E+00	0,00E+00	0,00E+00	0,00E+00	5,37E-02	ND	ND	ND	ND	ND	ND	ND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Materials for recycling (kg/FU)	0,00E+00	0,00E+00	6,67E-01	0,00E+00	8,37E-01	ND	ND	ND	ND	ND	ND	ND	0,00E+00	0,00E+00	1,38E+01	0,00E+00	0,00E+00
Materials for energy recovery (kg/FU)	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	0,00E+00	0,00E+00	0,00E+00	0,00E+00
Exported energy (MJ/FU)	0,00E+00	0,00E+00	4,70E-01	0,00E+00	7,05E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	0,00E+00	0,00E+00	2,86E+01	0,00E+00

9 IMPACT CATEGORIES ADDITIONAL TO EN 15804

		Production			Construction process stage		Use stage							End-of-life stage				
		A1 Raw material	A2 Transport	A3 manufacturing	A4 Transport	A5 Installation	B1 Use	B2 Maintenance	B3 Repair	B4 Replacement	B5 Refurbishment	B6 Operational energy use	B7 Operational water use	C1 Deconstruction / demolition	C2 Transport	C3 Waste processing	C4 Disposal	D Reuse, recovery, recycling
	PM (disease incidence)	1,195E-05	6,256E-08	1,739E-07	2,766E-08	3,791E-08	ND	ND	ND	ND	ND	ND	ND	0,00E+00	4,59E-08	1,94E-07	2,25E-08	-5,43E-06
	IRHH (kg U235 eq/FU)	8,132E+00	5,163E-03	1,205E-01	2,683E-03	7,857E-03	ND	ND	ND	ND	ND	ND	ND	0,00E+00	4,11E-03	7,09E-02	8,43E-03	-4,01E+00
	ETF (CTUe/FU)	2,141E+03	5,215E+00	1,442E+01	2,617E+00	2,917E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	4,05E+00	1,55E+01	2,26E+01	-2,27E+02
	HTCE (CTUh/FU)	5,615E-07	3,189E-10	1,868E-09	1,699E-10	8,239E-10	ND	ND	ND	ND	ND	ND	ND	0,00E+00	2,63E-10	1,98E-09	1,30E-09	-1,62E-07
	HTnCE (CTUh/FU)	6,150E-06	7,237E-09	3,414E-08	3,757E-09	6,781E-09	ND	ND	ND	ND	ND	ND	ND	0,00E+00	5,78E-09	8,90E-08	1,37E-08	-1,83E-06
	Land Use Related impacts (dimensionless)	3,920E+02	9,192E+00	7,988E+01	3,201E+00	4,521E+00	ND	ND	ND	ND	ND	ND	ND	0,00E+00	4,89E+00	1,94E+01	1,44E+00	-1,03E+02

HTCE = HUMAN TOXICITY – CANCER EFFECTS; HTnCE = HUMAN TOXICITY – NON-CANCER EFFECTS; ETF = ECOTOXICITY – FRESHWATER; (POTENTIAL COMPARATIVE TOXIC UNIT)

PM = PARTICULATE MATTER (POTENTIAL INCIDENCE OF DISEASE DUE TO PM EMISSIONS);

IRHH = IONIZING RADIATION – HUMAN HEALTH EFFECTS (POTENTIAL HUMAN EXPOSURE EFFICIENCY RELATIVE TO U235);

9.1 Environmental impact categories explained

	Global Warming Potential	The global warming potential of a gas refers to the total contribution to global warming resulting from the emission of one unit of that gas relative to one unit of the reference gas, carbon dioxide, which is assigned a value of 1. It is split up in 4: – Global Warming Potential total (GWP-total) which is the sum of GWP-fossil, GWP-biogenic and GWP-luluc – Global Warming Potential fossil fuels (GWP-fossil) : The global warming potential related to greenhouse gas (GHG) emissions to any media originating from oxidation and/or reduction of fossil fuels by means of their transformation or degradation (e.g. combustion, digestion, landfilling, etc). – Global Warming Potential biogenic (GWP-biogenic) : The global warming potential related to carbon emissions to air (CO₂, CO and CH₄) originating from the oxidation and/or reduction of aboveground biomass by means of its transformation or degradation (e.g. combustion, digestion, composting, landfilling) and CO₂ uptake from the atmosphere through photosynthesis during biomass growth - i.e. corresponding to the carbon content of products, biofuels or above ground plant residues such as litter and dead wood. – Global Warming Potential land use and land use change (GWP-luluc): The global warming potential related to carbon uptakes and emissions (CO₂, CO and CH₄) originating from carbon stock changes caused by land use change and land use. This sub-category includes biogenic carbon exchanges from deforestation, road construction or other soil activities (including soil carbon emissions).
	Ozone Depletion	Destruction of the stratospheric ozone layer which shields the earth from ultraviolet radiation harmful to life. This destruction of ozone is caused by the breakdown of certain chlorine and/or bromine containing compounds (chlorofluorocarbons or halons), Which break down when they reach the stratosphere and then catalytically destroy ozone molecules.
	Acidification potential	Acid depositions have negative impacts on natural ecosystems and the man-made environment incl. buildings. The main sources for emissions of acidifying substances are agriculture and fossil fuel combustion used for electricity production, heating and transport.
	Eutrophication potential	The potential to cause over-fertilization of water and soil, which can result in increased growth of biomass and following adverse effects. It is split up in 3: – Eutrophication potential - freshwater: The potential to cause over-fertilization of freshwater, which can result in increased growth of biomass and following adverse effects. – Eutrophication potential - marine: The potential to cause over-fertilization of marine water, which can result in increased growth of biomass and following adverse effects. – Eutrophication potential - terrestrial: The potential to cause over-fertilization of soil, which can result in increased growth of biomass and adverse effects.
	Photochemical ozone creation	Chemical reactions brought about by the light energy of the sun creating photochemical smog. The reaction of nitrogen oxides with hydrocarbons in the presence of sunlight to form ozone is an example of a photochemical reaction.
	Abiotic depletion potential for non-fossil resources	Consumption of non-renewable resources, thereby lowering their availability for future generations. Expressed in comparison to Antimony (Sb). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.
	Abiotic depletion potential for fossil resources	Measure for the depletion of fossil fuels such as oil, natural gas, and coal. The stock of the fossil fuels is formed by the total amount of fossil fuels, expressed in Megajoules (MJ). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.



	Ecotoxicity for aquatic fresh water	The impacts of chemical substances on ecosystems (freshwater). The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.
	Human toxicity (carcinogenic effects)	The impacts of chemical substances on human health via three parts of the environment: air, soil and water. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.
	Human toxicity (non-carcinogenic effects)	The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or there is limited experienced with the indicator.
	Particulate matter	Accounts for the adverse health effects on human health caused by emissions of Particulate Matter (PM) and its precursors (NOx, SOx, NH3)
	Resource depletion (water)	Accounts for water use related to local scarcity of water as freshwater is a scarce resource in some regions, while in others it is not. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or as there is limited experience with the indicator.
	Ionizing radiation - human health effects	This impact category deals mainly with the eventual impact on human health of low dose ionizing radiation of the nuclear fuel cycle. It does not consider effects due to possible nuclear accidents, occupational exposure nor due to radioactive waste disposal in underground facilities. Potential ionizing radiation from the soil, from radon and from some construction materials is also not measured by this indicator.
	Land use related impacts	The indicator is the "soil quality index" which is the result of an aggregation of following four aspects: – Biotic production – Erosion resistance – Mechanical filtration – Groundwater The aggregation is done based on a JRC model. The four aspects are quantified through the LANCA model for land use. The results of this environmental impact indicator shall be used with care as the uncertainties on these results are high or there is limited experience with the indicator.



10 DETAILS OF THE UNDERLYING SCENARIOS USED TO CALCULATE THE IMPACTS

10.1 A1 - raw material supply

This module takes into account the extraction and processing of all raw materials and energy which occur upstream to the studied manufacturing process.

10.2 A2 – transport to the manufacturer

The raw materials are transported to the manufacturing site.

10.3 A3 - manufacturing

This module considers the production process, including waste generated during production, as well as treatment of waste coming from raw materials packaging.

10.4 A4 – transport to the building site

Fuel type and consumption of vehicle or vehicle type used for transport	Truck 16-32 ton, EURO6
Distance	100 km
Capacity utilization (including empty returns)	50%
Bulk density of transported products	-
Volume capacity utilization factor	1

The default B-EPD PCR scenario of 100 km has been considered.



10.5 A5 – installation in the building

At the construction site, packaging materials become waste. The impact of packaging’s EOL treatment has been included in this module (until end-of-waste point). All other processes and materials are out of scope of this study,

Packaging waste type	Quantity, kg	Description
Wood	1.69E-02	RECYCLED
	7.38E-02	INCINERATED
	1.78E-01	LANDFILLED
Cardboard	2.03E-01	RECYCLED
	1.92E-02	INCINERATED
	1.49E-02	LANDFILLED
Plastic	2.36E-01	RECYCLED
	5.03E-01	INCINERATED
	2.86E-01	LANDFILLED



10.6 C: End of life

The following scenarios have been used for different materials to model end-of-life stage:

	Sorting	Landfill	Incineration	Recycling
Aluminium	100%	4%	-	96%
Rubbers		1%	99%	-
Plastic parts		10%	90%	-
Other metal parts		5%	-	95%
Coating, sealants, glues		-	100%	-

C1: No input is required to dismantle the window, as it is manual process.

C2: Default transport scenarios for Belgium have been considered for transport of waste.

C3: Only aluminium profiles & other metals are recycled at the end-of-life of the window.

C4: All plastic parts are either incinerated or landfilled at the end-of-life. Coatings, sealants, glues are incinerated with the low efficiency.

Module C2 – Transport to waste processing

Type of vehicle (truck/boat/etc.)	Fuel consumption (litres/km)	Distance (km)	Capacity utilization (%)	Density of products (kg/m ³)	Assumptions
Truck 16-32 t, EURO5	Default from ecoinvent 3.9.1	Sorting: 30 km Landfill: 50 km Incineration: 150 km Recycling: 200 km	50%	-	

End-of-life modules – C3 and C4

Parameter	Value (kg)
Wastes collected separately	14.4
Wastes collected as mixed construction waste	4.23
Waste for re-use	0
Waste for recycling	14.4
Waste for energy recovery	0
Waste for final disposal	4.23

10.7 D – Benefits and loads beyond the system boundaries

The following waste streams are considered in module D: aluminium, other metal parts, rubbers, coatings, sealants as well as the packaging of the window systems sent to treatment in A5 stage.

QUANTITATIVE DESCRIPTION OF THE LOADS BEYOND THE SYSTEM BOUNDARIES	Sorting & pressing of 12.6 kg of aluminium for recycling Sorting & pressing of 1.25 kg of metals for recycling Sorting & recycling of 0.23 kg of plastic waste (packaging) Sorting & shredding of 0.017 kg of wood waste (packaging) Sorting of 0.203 kg of paper waste (packaging)
QUANTITATIVE DESCRIPTION OF THE BENEFITS BEYOND THE SYSTEM BOUNDARIES	Avoided production of 12.6 kg primary aluminium Avoided production of 1.25 kg primary brass Avoided production of 0.23 kg primary PE (packaging) Avoided use of 0.017 kg of wood (packaging)



Avoided production of 0.203 kg of paper (packaging)

Avoided production of 32.5 MJ of heat using natural gas

Avoided production of 16.2 MJ of Belgian electricity mix



11 RELEASE OF DANGEROUS SUBSTANCES TO INDOOR AIR, SOIL AND WATER DURING THE USE STAGE

11.1 Indoor air

This product contributes to healthy indoor air quality by being aligned with the airtightness class C5 according to EN 1026 EN12207.

11.2 Soil and water

This product does not claim any performance in this regard.

12 DEMONSTRATION OF VERIFICATION

EN 15804+A2 serves as the core PCR

Independent verification of the environmental declaration and data according to standard EN ISO 14025:2010

Internal ☐ External ☒

Third party verifier:

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13 TECHNICAL INFORMATION FOR SCENARIO DEVELOPMENT

Parts of the installation	quantity	Description
Processes necessary for the installation of the product	<i>1 m²</i>	GLAZING, DOUBLE OR TRIPPLE, DEPENDING ON THE END-CLIENT OR THE REQUIRED U VALUES
Fixation & joining materials	<i>12</i>	STEEL WALL ANCHORS
Material losses	<i>0 kg</i>	BOTH WINDOW SYSTEM AND GLAZING ARE MADE TO SIZE



14 APPLICATION UNIT

This paragraph gives information on the applied product and how the reference flow and table with impacts relate to different applications.

STAR 75 window systems have only 1 application.

Results presented in this EPD are for 1 m² of a window system with the size < 2.3 m².

Different dimensions are possible, and window systems are made to measure, depending on the requirements of the customer and limited by the glass supplier.

To calculate the environmental impact of a window system with a different surface area:

- Results must be multiplied by the surface area (in m²) of the window system in question.

For example, if window system is 1.5 m², results presented in this EPD have to be multiplied by 1.5.

The **ratio between application and the reference** flow in the EPD is, therefore **1**.



15 ADDITIONAL INFORMATION ON REVERSIBILITY

Description	Type of fixing	Level of reversibility	Simplicity of disassembly	Speed of disassembly	Ease of handling (size and weight)	Robustness of material (material resistance to disassembly)	Comment
Outer or inner wall of a residential or commercial building	Steel wall anchors for mechanical fixing	Reversible connections with light repairable damage	Simple – requires the use of specific though common tools	Speed of disassembly varies from quick to slow depending on element dimensions	Easy to manipulate (by hand (small size and limited weight): one worker should be sufficient	The material resists well during disassembly	

16 BIBLIOGRAPHY

ISO 14040:2006: Environmental Management-Life Cycle Assessment-Principles and framework.
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NBN EN 15804+A2:2019
B-EPD PCR

General information

Owner of the EPD, Responsible for the data,
LCA and information



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Carbon systems v



Verifier

Bob Roijen & Agnes Schuurmans
SGS INTRON B.V.,
Date of verification: 3.06.2025
External independent verification of the declaration and data
according to EN ISO 14025 and relevant PCR documents



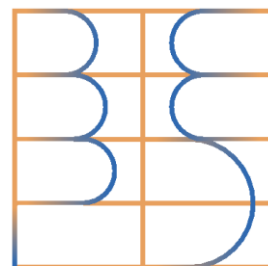
Comparing EPDs is not possible unless they are conform to the same PCR and taking into account the building context.
The program operator cannot be held responsible for the information supplied by the owner of the EPD nor LCA practitioner



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